INCH - POUND

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MILITARY SPECIFICATION

IGNITER, INTEGRAL, M56

This specification is approved for use by the U.S. Army Missile Command, Department *of* the Army, and is available for use by all Departments and Agencies of the Department *of* Defense.

1. SCOPE

1.1 scope. This specification covers special requirements not normally required by the item drawings, and provides the quality assurance provisions for the fabrication of parts, loading, assembling and packing for one type of integral igniter (containing an ignition transmission line) designated as Igniter, Integral, M56 for use in Rocket, HE, 66 mm, Anti-Tank, M72.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-P-2496	-	Primer, Percussion, M29A1 Parts and
	-	Loading, Assembling and Packing
MIL-I-45607	-	Inspection Equipment, Acquisition,
	-	Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1168	-	Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks l *re* available from the Standardization Documents Order Desk, Building 40, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

10048661	-	Carton, Packing, Ammunition, for Integral Igniter, Motor, Rocket: M56
10048662	-	BOX, Packing, Ammunition, for Integral Igniter, Motor, Rocket: M56
10048775	-	Integral Igniter, Motor, Rocket: M56 Loading Assembly
10048796	-	Housing, Primer
10048798	-	Flash Tube, Integral Igniter, Motor, Rocket: M56
10048799	-	Primer Block and Body Assembly, Igniter Motor, Rocket: M56
10048801	-	Line, Transmission, Ignition

(Unless otherwise indicated, requests for copies of Government drawings shall be made to the contracting officer.)

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PUBLICATIONS

ARMY

ET-10048775 - Equipment Tabulation

(Unless otherwise indicated, requests for copies of Government publications shall be made to the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.1.

3.2 Materials and parts. Materials and parts shall be in accordance with applicable drawings and specifications.

3.3 Assembly. The loaded igniter shall comply with all requirements specified on Drawing 10048775 and with all requirements specified in the applicable specifications.

3.4. Flash tube stress and crack resistance test. The flash tube (Drawing 10048798) shall not crack when bent at an angle of 180 degrees and subjected to a stress-cracking reagent environment at a temperature of 71° Celsius (C) (minimum) for a period of 24 hours as specified in 4.7.1 (see 4.6.1.1, 4.6.2.2 and 4.7.1).

3.5 Identification of molded parts. The molds used in the production of molded parts shall leave an identifying mark on the surface of the molded part to identify the mold from which it came.

3.6 Moisture content of the black Powder. The moisture content of the black powder at the time and place of loading shall not exceed 0.50 percent (see 4.6.2.3 and 4.7.2).

3.7 Waterproofness.

3.7.1 Parts assembly. The primer block and body assembly shall withstand an internal air pressure of 10 pounds (force) per square inch **(lbf/in^2)** gage (see 4.6.1.2 and 4.7.3).

3.7.2 Complete igniter. The complete igniter shall be waterproof as evidenced by functioning on the first attempt with ignition delay (see 3.8.1) not greater than 40 milliseconds (ms) after being submerged in water at a

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depth of 3 inches for a minimum of 24 hours with the water temperature maintained at 27°C (see 4.6.1.3 and 4.7.4).

3.8 Static functioning.

3.8.1 Ignition delay time (see 6.5.1). The ignition delay time shall be not greater than 40 ms when the igniter is operated at **-40°C.** The ignition delay the shall be not greater than 18 ms when the igniter is operated at 60°C (see 4.6.1.4 and 4.7.5).

3.8.2 Ignition time (see 6.5.2). The ignition time at **-40°C** shall be not less that 4.55 ms and shall be not greater than 12.8 ma, and at 60°C shall be not less than 2.4 ma, and shall be not greater than 9.6 ms (see 4.6.1.4 and 4.7.5).

3.8.3 Peak pressure (see 6.5.3). The peak pressure developed by the igniter shall be not greater than 840 **lbf/in^2** and shall be not less than 430 **lbf/in^2** when the igniter is operated as specified in 4.7.5 at -40°C. The peak pressure shall be not greater than 890 **lbf/in^2** and shall be not less than 440 **lbf/in^2** when the igniter is operated as specified in 4.7.5 at **60°C** (see 4.6.1.4 and 4.7.5).

3.9 Workmanship. The igniter shall be free of dirt, grease, black powder on exterior surfaces, wrinkled body lids or barriers, excessive adhesive, burrs and sharp edges or foreign material that may impair functioning or cause premature function. The cleaning method shall not be injurious to the igniter, nor shall the igniter be contaminated by the cleaning agents used.

3.10810 Check test for primer deterioration. The contracting *officer will* arrange to have the igniter contractor reimbursed for expenses incurred in the performance of the check test *for* primer deterioration when the contractor is directed to perform the test (see 4.6.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except *as* otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this

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specification shall become a part of the contractor's overall inspection system or quality program. The absence absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that l ll products or supplies submitted to the Government for acceptance comply with all requirements *of* the contract. Sampling inspection, as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.1, 4.5.2 and 4.6 below.

4.3.1 Lot formation. The term "lot-, as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity *of* units from which the sample was drawn and shall not be construed to represent any **prior or subsequent quantities presented for inspection. Homogeneity shall be** considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawings. specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain:

a. Primers from not more than two lots from one manufacturer

b. Black powder from not more than two lots from one manufacturer

Lots shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

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4.3.2 Submission of product . At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification

b. Number of units of product inspected

c. Results obtained for all inspections performed

d. Drawing, specification number and date, together with an identification and date of changes

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents

f. Number of items in the lot

g. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.3.3 Government verification. Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to ensure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified “critical" be accepted solely on the basis of the contractor's records.

4.4 First article inspection.

4.4.1 First article inspection. First article inspection shall be conducted on the first article sample (see 3.1) and also as specified by the procuring activity (see 6.2). If a lot should fail a first article inspection, no further lot will be accepted until the supplier has presented sufficient evidence to show that the condition which caused the failure has

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been corrected. The first article inspection shall consist of all tests of this specification (ace 4.5 and table I). Quantities to be inspected are to be specified by the procurement document (see 6.2).

4.4.2 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provision of 4.5. The first article sample shall consist of:

a. 15 sets of individual component parts

b. 10 parts assemblies

c. 50 flash tubes

d. 130 complete igniters

All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials (see 4.3.2.e). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government (see 6.2).

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table 1.

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TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph	Method paragraph
Flash tube stress-crack resistance	3.4	4.6.1.1	4.6.2.2, 4.7.1
Marking of molded parts	3.5	Visual	Visual
Moisture content of black powder	3.6	4.6.2.3	4.7.2
Waterproofness of parts	3.7.1	4.6.1.2	4.7.3
Waterproofness of complete assembly	3.7.2	4.6.1.3	4.7.4
Ignition delay time	3.8.1	4.6.1.4	4.7.5
Ignition time	3.8.2	4.6.1.4	4.7.5
Peak pressure	3.8.3	4.6.1.4	4.7.5
Workmanship	3.9	Visual	Visual
Check test for primer deterioration	3.10	4.6.2.1	------

4.5.1 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and Interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection *of* the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-229. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.5.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with tables II and III and all provisions of this specification. Equipment necessary for the performance of the inspections shall be in accordance with 4.6.3.

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TABLE II. Sampling plan A.

Lot Size	Sample Size
2 to 13	100%
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 and over	102
In all cases: Acceptance number is ZERO Rejection number is ONE	

TABLE III. Sampling plan B.

Lot Size	Sample Size
2 to 32	100%
33 to 280	32
281 to 500	48
501 to 3200	73
3201 tO 10,000	86
10,001 to 35,000	108
35,001 to 150,000	123
150,001 to 500,000	156
500,001 and over	189
In all cases: Acceptance number is ZERO Rejection number is ONE	

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4.5.2.1 Housing, Primer (see drawing 10048796 covering a detail of drawing 10048799 and table IV).

TABLE IV. Classification of defects, housing, primer.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	(Table III. Sampling plan B.)	
101.	Concentricity of hole in bottom cavity with large outside diameter	Gage
102.	Diameter of cavity	Gage
103.	Diameter of hole	Gage
104.	Depth of cavity	Gage
105.	Concentricity of diameter of cavity with large outside diameter	Gage

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4.5.2.2 Primer block and body assembly, igniter, motor, rocket: M56 (see 3.4 and drawing 10048799 covering a detail of drawing 10048775 and table V-

TABLE V. Classification of defects, primer block and body assembly, igniter, motor, rocket: M56.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	(Table III. Sampling plan B.)	
101.	Height of flash tube from bottom of igniter cup, max	Gage
102.	Inside diameter of flash tube at igniter cup end, necked below min	Gage
103.	Large outside diameter of igniter cup	Gage
104.	Outside diameter of primer holder section	Gage
105.	Height of primer holder section	Gage
106.	Depth of charge cavity in igniter cup	Gage
107.	Tube kinked, cracked or assembly damaged	Visual
108.	Total length	Gage
109.	Outside diameter along entire length of recessed motor throat engagement diameter, min	Gage
110.	Angle between primer block and flash tube, max	Gage

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4.5.2.3 Line, transmission, ignition (see drawing 10048801 covering a detail of drawing 10048775 and table VI).

TABLE VI. Classification of defects, line, transmission, ignition.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	(Table II. Sampling plan A.)	
101.	Explosive missing or incomplete	Visual

4.5.2.4 Integral igniter, motor, rocket: M56, loading assembly, prior to assembly *of* primer and black powder (see drawing 10048775 and table VII) .

TABLE VII. Classification of defects, integral igniter, motor, rocket: M56, loading assembly, prior to assembly *of* primer and black powder.

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4.5.2.5 Integral igniter, motor, rocket: M56, loading assembly, prior to heat seal of lid (see drawing 10048775 and table VIII).

TABLE VIII. Classification of defects, integral igniter, motor, rocket: M56, loading assembly, prior to heat seal of lid.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	(Table II. Sampling plan A.)	
101.	Weight of black powder	Balance

4.5.2.6 Integral igniter, motor, rocket: M56, loading assembly (see drawing 10048775 and table IX).

TABLE IX. Classification of defects, integral igniter, motor, rocket: M56, loading assembly.

Categories	Defects	Method of Inspection
Critical:		
1.	Primer above flush	Gage
2.	Holes in lid or parts loose to the extent that the explosive charge may leak out or draw moisture	Visual
Major:	(Table II. Sampling plan A.)	
101.	Igniter improperly assembled	Visual

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4.5.2.7 Carton, packing, ammunition, for integral igniter, motor, rocket: M56 (see drawing 10048661 and table X).

TABLE X. Classification of defects, carton, packing, ammunition for integral igniter, motor, rocket: M56.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	(Table II. Sampling plan A.)	
101.	Contents loose	Manual
102.	Carton improperly sealed	Visual

4.5.2.8 Box, packing, ammunition, for integral igniter, motor, rocket: M56 (see drawing 10048662 and table XI).

TABLE XI. Classification of defects, box, packing, ammunition, for integral igniter, motor, rocket: M56.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:	(Table III. Sampling Plan B.)	
101.	Box broken or damaged	Visual
102.	Strapping missing or loose	Visual/Manual
103.	Marking missing, incomplete, incorrect or illegible	Visual
104.	Moisture barrier (when required) not completely sealed	Visual/Manual
105.	Moisture barrier (when required) torn or punctured	Visual

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4.6 Testing.

4.6.1 First article (see 4.4.2). The samples submitted as the first article quantity shall be tested as specified herein after being examined for all drawing dimensions and drawing requirements.

4.6.1.1 Flash tube stress-crack resistance test (see 3.4) Major defect. A sample of 50 flash tubes shall be selected from each lot and tested as specified in 4.7.1 using test equipment specified in 4.6.3.

4.6.1.2 Waterproofness of the parts assembly (see 3.7.1). The 10 parts assemblies shall be subjected to the test specified in 4.7.3 using test equipment specified in 4.6.3.

4.6.1.3 Waterproofness *of* the complete igniter (see 3.7.2) Duds; Ignition delay time greater than the specified limit; Ignition delay time greater than 40 ms. Fifty of the complete igniters shall be tested as specified in 4.7.4 at the specified temperature using the test equipment specified in 4.6.3.

4.6.1.4 Static functioning (see 3.8); Duds; Ignition delay time greater than requirement at -40°C; Ignition delay time greater than 40 ms at **-40°C;** Ignition delay time greater than requirement at 60°C; Ignition delay time greater than 18 ms at **60°C;** Ignition time at **-40°C;** Ignition time at **60°C;** Peak **pressure at -40°C and Peak pressure at 60°C.** Forty complete igniters shall be tested at **-40°C** and 40 complete igniters shall be tested at **60°C.** The test shall be performed as specified in 4.7.5 using teat equipment specified in 4.6.3. If one or more assemblies is classed defective, the lot shall be rejected.

4.6.2 Production.

4.6.2.1 Check test for possible deterioration of primers (see 3.10): Major defect. If the total time between original acceptance of the primer lot and the assembly of the primers into the igniter exceeds two years, or *if* the primers show evidence of having been subjected to adverse environmental conditions for any time, *however* brief, the primer lot shall be subjected to the check test for possible deterioration specified in MIL-P-2496. The primer lot shall be considered usable if the test results conform to the requirements of MIL-P-2496. When required, this test shall be performed by the contractor on primers selected by the inspector at the facility assembling the igniter. The test shall be conducted at Government expense without cost to the contractor who produced the primers or the contractor assembling the primers into the igniter, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling or storage of the primer lot while the primer lot was under the jurisdiction of either contractor.

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4.6.2.2 Flash tube stress-crack resistance test (see 3.4): Major defect. A sample of 50 flash tubes shall be selected from -each lot and tested as specified in 4.7.1 using test equipment specified in 4.6.3. If one or more flash tubes are classed defective, the lot shall be rejected.

4.6.2.3 Moisture content of the black powder (see 3.6): Major defect. A sample of approximately three grams of black powder shall be selected from each container when loading from the container is initiated and every four hours thereafter, until the contents of the container are depleted. The sample shall be divided into three approximately equal specimens and each specimen tested as specified in 4.7.2. If one or more specimens fail to conform to the requirement specified the quantity loaded from that container since the last test shall be rejected.

4.6.2.4 Waterproofness of the parts assembly (see 3.7.1): Major defect. A sample of 50 parts assemblies (drawing 10048799) shall be selected from each lot and tested as specified in 4.7.3 using test equipment specified in 4.6.3. If one or more assemblies is classed defective the lot shall be rejected.

4.6.2.5 Waterproofness of the complete igniters (see 3.7.2); Duds: Major defect; Ignition delay time greater than the specified limit: Major defect; Ignition delay time greater than 40 w: Critical defect. A sample of 50 complete igniters shall be selected from each lot and tested as specified in 4.7.4. using test equipment specified in 4.6.3. If one or more igniters fail to meet the requirements for ignition delay time or proves to be a dud, the lot shall be rejected.

4.6.2.6 Static functioning see 3.8}; Duds: Major defect; Ignition delay time greater than requirement at -40°C: Major defect; Ignition delay time greater than 40 ms at -40°C: Critical defect; Ignition delay time greater than requirement at 60°C: Major defect; Ignition delay time greater than 18 ma at 60°C: Critical defect; Ignition time at -40°C: Major defect; Ignition time at 60°C: Major defect; Peak pressure at -40°C: Major defect and peak pressure at 60°C: Major defect. A sample of 80 complete igniters shall be selected from each lot and tested as specified in 4.7.5 using test equipment specified in 4.6.3. Forty igniters shall be tested at -40°C, and 40 igniters shall be tested at 60°C. The lot shall be rejected if one or more igniters exceed the ignition delay time limit at either temperature or if one or more duds are observed. The lot shall also be rejected if the average ignition delay time at either temperature plus the product of 2.01 and the standard deviation (see 6.4) is greater than the specified requirement or if the average ignition time at either temperature ± the product of 2.01 and the standard deviation fails to conform to the specified requirements or if the average peak pressure at either temperature ± the product of 2.01 and the standard deviation fails to conform to the specified requirements.

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4.6.3 Inspection equipment. Equipment Tabulation (ET) number ET-10048775 identifies the inspection equipment required to perform the examinations and tests prescribed in this section.

4.6.3.1 Government rights to documentation. Inspection equipment drawings and lists provided and revised in accordance with the requirements of ET-10048775 may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.6.3.2 Supply and maintenance. Supply and maintenance of the equipment listed on ET-10048775 shall be in accordance with MIL-I-45607.

4.6.3.3 Government use of contractor's inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examinations or tests by the Government.

4.6.4 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of Drawings 10048661 and 10048662.

4.7 Test methods and procedures.

4.7.1 Flash tube stress-crack resistance test. The flash tube shall be conditioned to the required temperature for a minimum of 10 minutes prior to bending. Following the minimum conditioning period, each flash tube shall be uniformly and smoothly bent to the specified angle and placed in a fixture for holding the tube at the required angle. The entire bent exterior surface of the tube shall then be completely immersed in the stress-cracking reagent and while immersed shall be conditioned at the specified temperature. Following immersed conditioning for the required time, each flash tube shall be visually examined for cracks. The flash tube shall be classed defective if it fails to meet the specified requirement.

4.7.2 Moisture content of the black powder. The black powder specimen shall be accurately weighed and transferred to a tared weighing dish or watch glass and dried *for* five hours, uncovered, at 70°C to 75°C. The dish shall be covered, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percent moisture of the black powder.

4.7.3 Waterproofness of the parts assembly. The parts assembly shall be assembled to the specified test equipment, the assembly submerged in acetone or water containing a wetting agent, and the required internal air pressure applied. The assembly shall be classed defective if a steady stream

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of bubbles is emitted from any portion of the assembly. Bubbles formed by incomplete seals between the teat equipment and the assembly shall not be cause for classifying the assembly defective.

4.7.4 Waterproofness of the complete igniter. The complete igniter shall be placed under the specified depth of water, the temperature of which is 21°C to 32°C, for the required time. The igniter shall then be removed from the water, assembled to the test equipment, and operated. The igniter shall be classed defective if it fails to operate on the first attempt or if the ignition delay time is greater than the specified limit.

4.7.5 Static functioning. The igniter shall be conditioned at the required temperature (-40°C or 60°C, ± 2.8°C) for a minimum of six hours and operated between two and three minutes after being removed from the conditioning chamber. The test equipment shall be capable of obtaining two pressure time traces simultaneously. Averages of the two traces will be used in the final data reduction. If one instrumentation channel fails, the results obtained with the remaining charnel shall be considered absolute and adequate.

5. PACKAGING

5.1 Preservation. Preservation shall be either levels B or C. as specified below.

5.1.1 Level B preservation. The igniters shall be packaged in accordance with drawings 10048661 and 10048662 using the specified moisture barrier.

5.1.2 Level C preservation. The igniters shall be packaged in accordance with drawings 10048661 and 10048662 except that the moisture barrier may be omitted.

5.2 Packing. Packing for levels B and C shall be in accordance with drawing 10048662.

5.3 Marking. Markings shall be in accordance with those specified on drawings 10048661 and 10048662.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The igniter assembly covered by this specification is intended for use in a Rocket, HE, 66 mm, Anti-Tank, M72.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether a first article sample is required (see 3.1 and 4.4.2) and how many samples will be required (see 4.4.2).

d. Data cards. Data cards shall be prepared for each lot in accordance with MIL-STD-1167.

e. Applicable levels of preservation (see 5.1).

6.3 First article. When first article inspection is required, the items should be a first article sample. The first article should consist of:

a. 15 sets of individual component parts
b. 10 parts assemblies
c. 50 flash tubes
d. 130 complete igniters.

The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior government approval is presently appropriate for the pending contract.

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6.4 Calculation. The sample standard deviation shall be calculated using the following formula or its equivalent:

$$S = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n - 1}}$$

where:

s = sample standard deviation

X = numerical value for individual observations

X = average value for the sample

n = sample size.

6.5 Definitions.

6.5.1 Ignition delay time. The time interval from when the firing pin contacts the primer until the 25 psi pressure level is recorded on a pressure time trace.

6.5.2 Ignition time. Time interval from when the pressure time trace reaches 25 psi to the 90 percent peak pressure point.

6.5.3 Peak pressure. The highest point on the pressure time trace.

6.6 Subject term (keyboard) listing.

Black powder
Flash tubes
Igniter assembly
Primer, percussion

6.7 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this document.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing Activity:
Army - MI

Review Activity:
Army-AR

Project No. 1340-A488

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-60008B (MI)	25 February 1991

3. DOCUMENT TITLE

Igniter, Integral, M56

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	e. DATE SUBMITTED (YYMMDD)

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Commander J.S. Army Missile Command	(205) 876-6980	746-6980
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